

Loan Approval Predictor

Internship Project Report — Task 4

1. Introduction

Manually assessing loan applications is time-consuming and can be inconsistent across reviewers. This project builds a machine learning pipeline that predicts whether a loan application should be approved based on applicant details such as income, credit history, education, and property area, using a public Loan Prediction dataset.

2. Objective

The objective is to preprocess and engineer features from the Loan Prediction dataset, train and compare three classification models (Logistic Regression, Random Forest, and XGBoost), evaluate the best model, and expose it through both a reusable prediction module and an interactive Streamlit web application.

3. Dataset Description

The dataset contains 491 loan applications with a mix of categorical and numeric features, along with a binary target (Loan_Status) indicating whether the loan was approved. Several columns contain real missing values, reflecting a realistic data-quality scenario.

Feature	Description
Gender	Male / Female
Married	Applicant's marital status
Dependents	Number of dependents (0, 1, 2, 3+)
Education	Graduate / Not Graduate
Self_Employed	Whether the applicant is self-employed
ApplicantIncome	Applicant's monthly income
CoapplicantIncome	Co-applicant's monthly income
LoanAmount	Requested loan amount (in thousands)
Loan_Amount_Term	Loan repayment term, in days
Credit_History	1 = has credit history, 0 = does not
Property_Area	Urban / Semiurban / Rural
Loan_Status	Target: 1 = approved, 0 = rejected

4. Data Preprocessing

The non-predictive Loan_ID identifier was dropped. Missing values in categorical columns (Gender, Married, Dependents, Self_Employed, Credit_History) were imputed with the mode of each column, while missing values in numeric columns (LoanAmount, Loan_Amount_Term) were imputed with the median.

5. Feature Engineering

Beyond the raw features, four derived features were added to better capture loan affordability: TotalIncome (combined applicant and co-applicant income), LoanAmountToIncome (a debt-to-income style ratio), EMI (an approximate monthly repayment burden), and HasCoapplicant (a binary flag). Categorical variables were one-hot encoded, the target was label-encoded, and all numeric features were standardized after a stratified 80/20 train-test split.

6. Model Training

Three classification algorithms were trained and evaluated on the same preprocessed data: Logistic Regression, Random Forest (200 trees, max depth 6), and XGBoost (200 boosting rounds, max depth 4, learning rate 0.05). The model with the highest test accuracy was selected as the final model.

7. Results

Model	Test Accuracy
Logistic Regression (Best)	84.8%
Random Forest	83.8%
XGBoost	78.8%

Logistic Regression achieved the best test accuracy (84.8%) with an ROC-AUC of approximately 0.86, slightly outperforming Random Forest and clearly outperforming XGBoost on this relatively small dataset (491 records), where simpler linear models often generalize better than more complex ensemble methods.

8. Visualizations

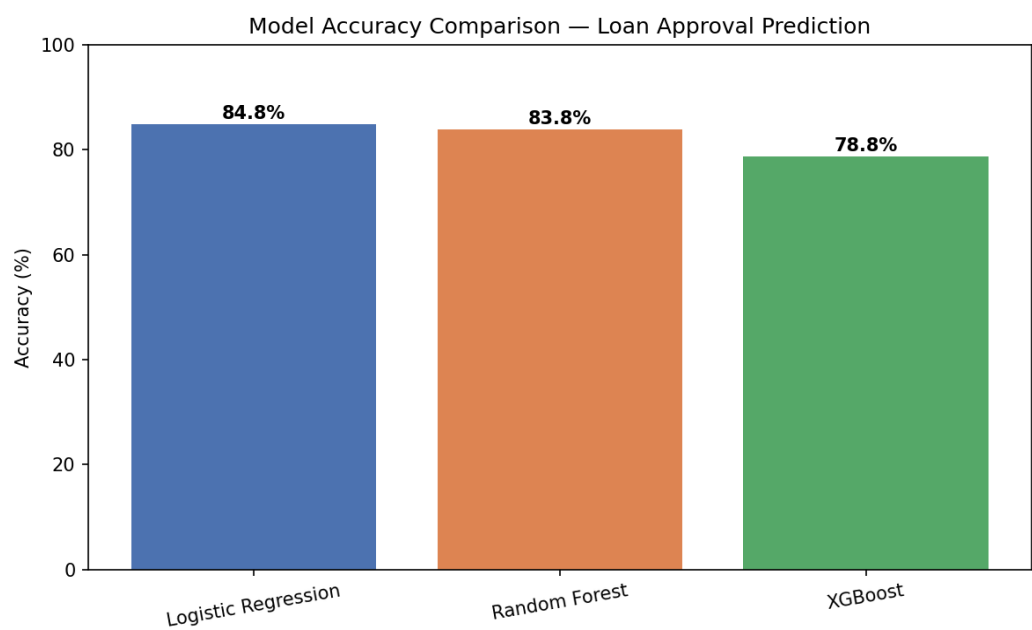


Figure 1: Model accuracy comparison across the three classifiers.

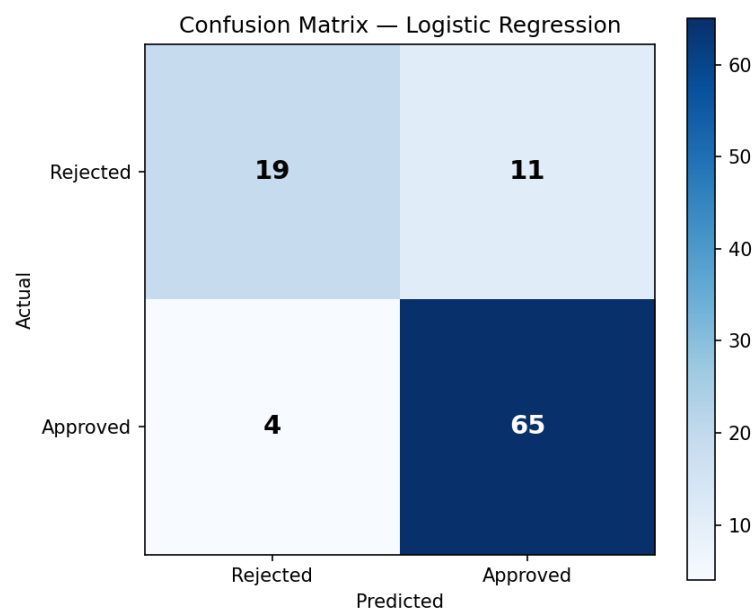


Figure 2: Confusion matrix of the best model (Logistic Regression) on the test set.

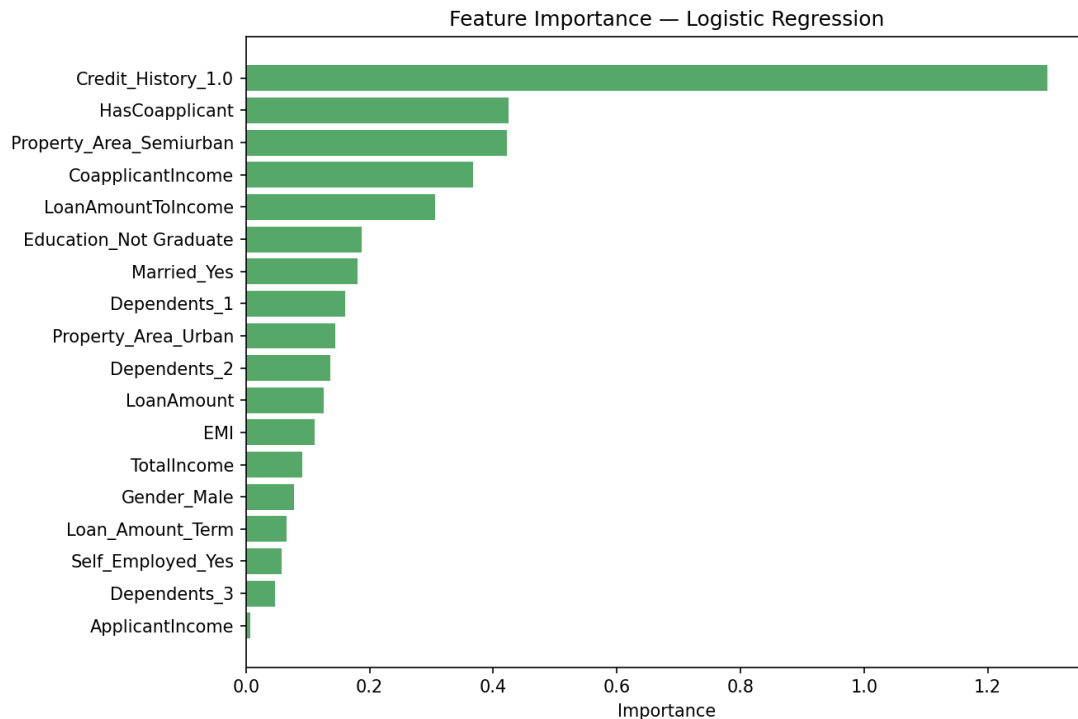


Figure 3: Feature importance (absolute coefficient magnitude) for the best model.

9. Prediction Interface & Web App

A reusable prediction module (`src/prediction.py`) loads the saved model artifact and predicts approval for new applicants given as a Python dictionary. As a bonus, a Streamlit web application (`app/streamlit_app.py`) provides an interactive form where a user can enter an applicant's details through a browser UI and instantly see the predicted decision along with the model's confidence score.

10. Conclusion

This project demonstrates a complete, reproducible loan-approval prediction pipeline: from raw data cleaning and feature engineering through model comparison to a saved, deployable model artifact with both a programmatic interface and an interactive web app. Credit History and Total Income emerged as the most influential predictors of loan approval, consistent with real-world lending practice.

11. Future Improvements

Potential enhancements include hyperparameter tuning via grid/randomized search, k-fold cross-validation for more robust accuracy estimates, explicit handling of class imbalance (e.g. SMOTE or class weighting), SHAP-based per-prediction explanations, and deploying the Streamlit app publicly for live demonstrations.